Randy Dyer

rbdyer3@gmail.com randydyer.dev randyer

WORK EXPERIENCE

Software Engineer, Space Age Electronics Jan 2025 – Sep 2026

- Developed touchscreen annunciator for fire and life safety using Electron.js to integrate with fire panels and visualize real-time device statuses via serial connection.
- Built admin version of annunciator with drag-and-drop editor that generates configuration files defining UI layout and device point-to-point connections, enabling rapid deployment for custom buildings.
- Presented live product demos to executive stakeholders and investors.
- Developing tablet-based UI and Voice over IP network infrastructure for an emergency two-way communication system.
- Built 3D configurator integrated into Epicor ERP system for positioning electrical components in custom fire safety cabinets and automatically generating AutoCAD drawings, streamlining the design workflow.
- Built a Power Automate workflow to route engineering requests to Microsoft Planner with automated submission and completion notifications.
- Created Power BI dashboard tracking key performance metrics including average task turnaround time.

App Developer and Web Designer, Alton Therapeutic Massage Jul 2024 – Jan 2025

- Designed and implemented a full-stack React web application hosted on AWS Amplify to centralize client information such as contact details, health records, symptoms, and appointment notes.
- Increased client inquiries by 70% through the creation of the company website.
- Stored sensitive data (PHI) in encrypted Amazon RDS and ensured secure user authentication using Cognito.
- Streamlined new client onboarding by digitizing the intake form process, reducing initial consultation time.

Mobile Developer, Student Success Application Jul 2023 – Nov 2023

- Designed and implemented a React Native frontend for a virtual student handbook, improving content accessibility for Academic Support, Career Development, Culture & Diversity and Wellness resources.

PROJECTS

Beyond the Visual Application, Major Qualifying Project Aug 2023 – Mar 2024

- Worked with a team to develop the Worcester Art Museum's mobile app using React Native, mongoDB and Firebase.
- Developed and tested an indoor positional system (IPS) using bluetooth beacons and trilateration techniques to locate and navigate individuals through exhibits.
- Added accessibility labels to UI for screen readers and added an audio player for art piece descriptions.
- Conducted user testing with both visually impaired and non-visually impaired individuals on all app features.

Hospital Service Application, Brigham and Women's Hospital Mar 2023 – May 2023

- Led back-end development for a team of 10, managing the AWS PostgreSQL RDS and associated Java code.
- Facilitated the integration of features with the front-end such as submitting and receiving service requests, hospital navigation, and admin privileges for map editing and employee updates.
- Utilized Agile Methodology and Software Design patterns throughout the creation of the app.

Water Conservation Outreach, Board of Water supply Jan 2023 – Feb 2023

- Collected and analyzed 168 survey responses to identify water conservation educational content and determine optimal platforms for reaching individuals across age groups.
- Created a memory matching game with HTML, CSS, and JavaScript to teach children water conservation tips.
- Conducted user testing through local elementary schools and obtained survey feedback from teachers.

SKILLS

Programming Languages: Typescript, JavaScript, Java, Python, SQL, noSQL, C, C++

Development: Agile, Git/GitHub, Node.js, Electron.js, AWS (EC2, RDS, Lamda, VPC, Cognito, Amplify, API Gateway), REST API, PostgreSQL, mongoDB, Pandas, NumPy, React/React Native, macOS, Windows, Linux/Ubuntu

Software/Tools: VSCode, Jira, Figma, AutoCAD, SolidWorks, Microsoft Power Automate

EDUCATION

Worcester Polytechnic Institute, Bachelor of Science in Computer Science May 2024 | Worcester, MA

PERSONAL INTERESTS

Golf, Pickleball, Basketball, Hiking, Cooking, Game Development, Digital Music Creation